Mapathon 2025 – Call for Participation

The Mapathon 2025, organised by the Society of Geoinformatics Engineers at the Institute of Remote Sensing (IRS), Anna University, in partnership with iTNT (Tamil Nadu Technology Hub), serves as an advanced collaborative contest uniting academia, researchers, and start-ups to solve leading geospatial challenges through automation and innovative mapping techniques.

The initiative seeks to encourage participants to develop **automated, scalable, reproducible, and commercially viable and real-time update approaches** that can transform the way spatial datasets are analysed, interpreted, and documented.

By leveraging **open-source tools, remote sensing, GIS, and advanced data analytics**, the Mapathon aims to generate impactful solutions in the domains of:

- Environmental monitoring
- Sustainable resource management
- Coastal and disaster risk studies
- Geospatial data cataloguing and knowledge sharing

Problem Statement 1: Automated Change Mapping

Develop automated methodologies for automated land use and land cover (LULC) change detection using multi-temporal raster datasets and a defined classification scheme.

Data Provided:

- LULC maps of multiple themes and their corresponding historical satellite data
- Current-period satellite data of the same area
- A defined classification scheme
- Raster datasets for scalable and accurate detection of spatial and temporal changes

Participants will analyse both past and current thematic LULC maps alongside the corresponding satellite data to design scalable, reproducible approaches for spatial and temporal change identification.

Problem Statement 2: Automated Water Spread Detection and Volume Estimation

Devise efficient techniques for **automated extraction of water spread areas** from multi-seasonal satellite imagery. The challenge

- Focus on enhancing accuracy and efficiency in water body mapping
- Estimate seasonal variations in water spread and approximate storage/volume changes
- Contribute towards **sustainable water resource management.**

Problem Statement 3: Automated Shoreline Analysis

Develop tools for **automated shoreline estimation and monitoring** using multi-sensor satellite imagery and bathymetric data.

- Detect and quantify shoreline changes over time.
- Support coastal zone management, climate impact studies, and disaster risk reduction.
- Generate accurate water delineation reports.

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Problem Statement 4: Comprehensive Cataloguing and Documentation of Map Datasets

Create frameworks for systematic cataloguing, integration, and documentation of diverse geospatial datasets across time and space.

- Integrate and ensure the continuity of coastal parameters like HTL at area/block/district levels with multi-temporal datasets.
- Ensure interoperability, accessibility, and reusability of spatial data
- Provide solutions for long-term geospatial dataset management for research and decision-making

Categories of Participation

- **Academic Institutions**
  - Student teams: Minimum 3 and maximum 4 members
  - Each team must have one faculty mentor
- **Start-ups**
  - Dedicated participation track
  - Additional support through capacity-building initiatives

Prizes

- **First Prize – ₹50,000 + Trophy**
- **Second Prize – ₹25,000 + Trophy**
- **Third Prize – ₹10,000 + Trophy**

(Three prizes will be awarded separately in each category – Academic Institutions and Start-ups.)

Special Benefits:

- Start-ups will receive **free access to iTNT's bootcamp**
- Winning teams and start-ups may sign an **MoU with IRS** for involvement in **research and sponsored projects**

Important Dates

- **November 30, 2025** – Deadline for Registration

- **December 30, 2025** – Stage 1 Selection Announcement
- **January 30, 2026** – Final Submission Deadline

Roles and Responsibilities

Institute of Remote Sensing (IRS), Anna University

- Conceptualise and frame the problem statements.
- Provide curated datasets, classification schemes, and technical resources.
- Offer academic mentorship and guidance to participants.
- Lead the **technical evaluation** of submissions through subject experts.
- Facilitate research collaborations and sponsored projects through MoUs with winning teams and start-ups.

iTNT(iTNT (Tamil Nadu Technology Hub)**, Tamil Nadu**

- Serve as the **marketing and outreach partner**, ensuring wider participation and visibility across India.
- Support **data dissemination, result compilation, and documentation** during the competition.
- Participate in the **jury process** along with the IRS, with a focus on evaluating **commercial viability, scalability, and real-time implementation potential** of proposed solutions.
- Act as a **media partner and facilitator**, ensuring seamless communication and outreach of information from the initial announcement stage through to the finalisation of results strictly adhering to the prescribed timelines and schedules.
- Provide dedicated support for **start-ups**, including mentorship, incubation, and access to iTNT's bootcamps.
- Facilitate pathways for scaling innovative solutions towards **industry adoption and commercialisation**.

Mapathon 2025 provides a unique opportunity for students, researchers, and start-ups to showcase their skills in geospatial analytics and contribute towards addressing critical environmental and societal challenges. By combining academic rigour with industry-focused innovation, the initiative seeks to build a stronger geospatial ecosystem in Tamil Nadu and beyond.